Matrix Operations and Inverse Matrices

Sections 2.4, 2.5 and 2.6 explained simply

2.4 Rules for matrix operations · 2.5 Inverse matrices · 2.6 Elimination = Factorization: A = LU

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The big picture: 2.4, 2.5, 2.6

2.4 Multiply

$$\begin{pmatrix} 1 & 1 \\ 2 & -1 \end{pmatrix} \times \begin{pmatrix} 2 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 5 & 6 \\ 1 & 0 \end{pmatrix}$$

Combine two matrices: rows times columns. Order matters.

2.5 Undo

$$\begin{pmatrix} 1 & 2 \\ 3 & 7 \end{pmatrix} \times \begin{pmatrix} 7 & -2 \\ -3 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

The inverse A^{-1} cancels A, like dividing by a number. Not every matrix has one.

2.6 Split

$$\begin{pmatrix} 1 & 1 \\ 3 & 5 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix} \times \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$$

A = LU breaks A into two simple triangular pieces, which makes solving easier.

Think of a matrix as a machine that changes numbers. We learn to chain machines (2.4), undo them (2.5) and take them apart (2.6).

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2.4 What is a matrix?

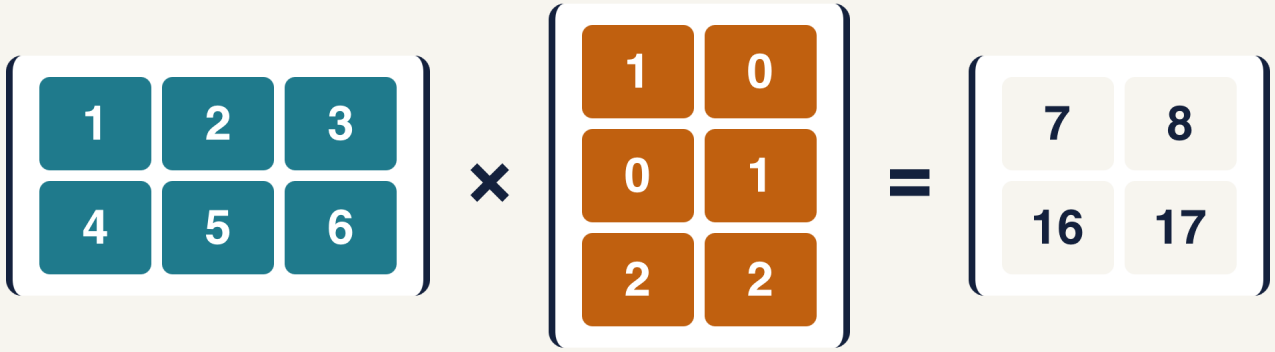


A matrix is a rectangle of numbers. This one has 2 rows and 3 columns, so it is a 2×3 matrix. The orange entry sits in row 2, column 3.

Same-size matrices add entry by entry:

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} + \begin{bmatrix} 2 & 2 \\ 4 & 4 \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ 7 & 8 \end{bmatrix}$$

2.4 The size rule

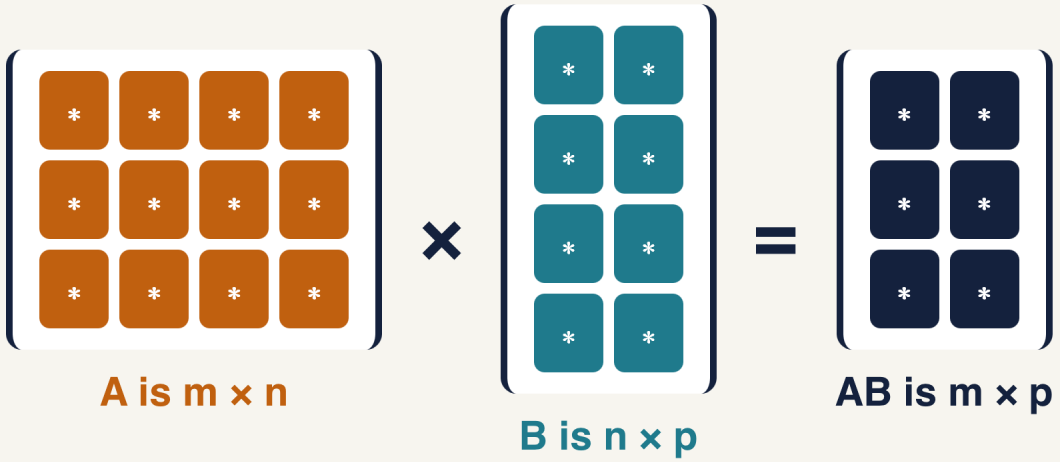


A is 2×3 and B is 3×2. The inner numbers (3 and 3) match, so we can multiply.

The outer numbers (2 and 2) give the size of the answer: 2×2.

If the inner numbers differ, the product does not exist.

2.4 Multiplying matrices: the size rule



The inner number n must match: columns of A = rows of B (here 4 and 4).

The outer numbers m and p give the size of the answer.

Each entry of AB is a dot product, and AB takes m·n·p multiplications.

2.4 All four entries, step by step

$$\begin{bmatrix} 1 & 1 \\ 2 & -1 \end{bmatrix} \times \begin{bmatrix} 2 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 5 & 6 \\ 1 & 0 \end{bmatrix}$$

Row 1 · Column 1

$1 \cdot 2 + 1 \cdot 3 = 5$

Row 1 · Column 2

$1 \cdot 2 + 1 \cdot 4 = 6$

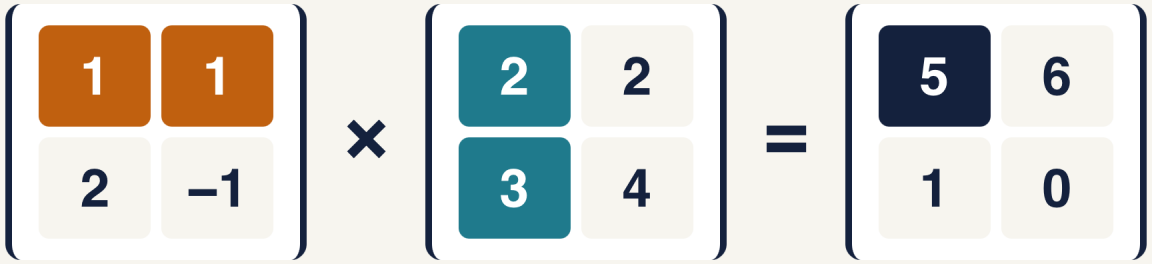
Row 2 · Column 1

$2 \cdot 2 + (-1) \cdot 3 = 1$

Row 2 · Column 2

$2 \cdot 2 + (-1) \cdot 4 = 0$

Visual example: one entry of AB



Row 1 of A (orange) · column 1 of B (teal) = 1·2 + 1·3 = 5. That is the top-left entry of AB.

Repeat for every row and column pair: 4 dot products, 8 multiplications.

2.4 Four ways to see AB

1. Rows × columns



One dot product for each entry.

2. A × each column of B



Each column of AB mixes the columns of A.

3. Each row of A × B



Each row of AB mixes the rows of B.

4. Columns × rows



Column × row is a full matrix. Add them all.

The rules of matrix multiplication

Always true

A(B + C) = AB + AC

(A + B)C = AC + BC

A(BC) = (AB)C, so parentheses are not needed.

Usually false

AB ≠ BA.

Order matters: the matrices do not commute.

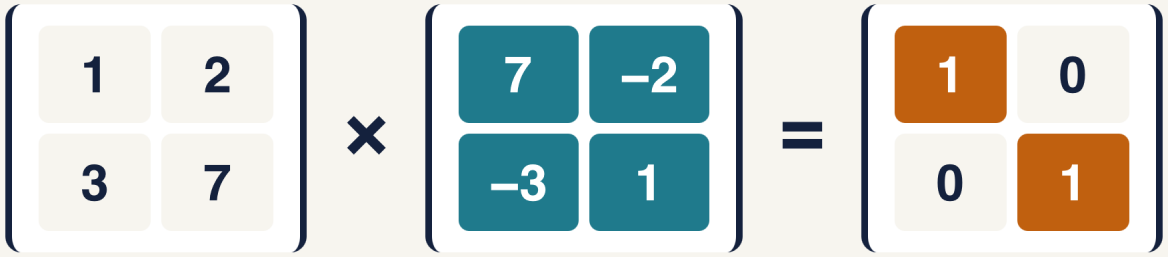
Example: [0 0; 1 0] times [0 1; 0 0] differs from the reverse order.

Block multiplication: cut matrices into blocks and multiply the blocks like numbers, keeping A’s in front of B’s.

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2.5 The inverse is an undo button

Numbers: $5 \times 1/5 = 1$ **Matrices: $A \times A^{-1} = I$**



What it means

I is the identity matrix, the matrix version of 1. It changes nothing.

A^{-1} cancels A, in either order.

Why we care

$Ax = b$ is solved by $x = A^{-1}b$.

Example: $b = (3, 10)$ gives $x = (7 \cdot 3 - 2 \cdot 10, -3 \cdot 3 + 10) = (1, 1)$.

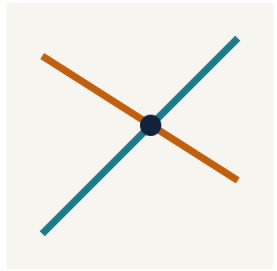
2.5 When there is no inverse

Elimination shows it

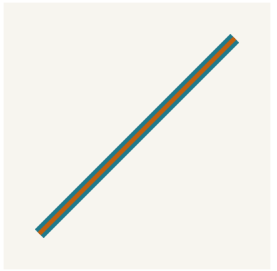
$$\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \xrightarrow{\text{row 2} - 2\times\text{row 1}} \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}$$

Row 2 is 2 × row 1, so elimination leaves a row of zeros: only 1 pivot, but we need 2.

Geometry shows it



Invertible: one crossing point



Singular: same line, endless answers

Fewer pivots

Fewer than n pivots means singular.

A hidden solution

Ax = 0 has x = (2, –1), not just 0.

Determinant zero

ad – bc = 1·4 – 2·2 = 0.

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Inverse matrices

A^{-1} undoes A: $A^{-1}A = I$ and $AA^{-1} = I$. Only square matrices can have one.

A is invertible if…

Elimination gives n pivots.

The determinant is not zero.

Ax = 0 only has x = 0.

Useful rules

$(AB)^{-1} = B^{-1}A^{-1}$: socks then shoes, undo in reverse.

2×2: swap a and d, negate b and c, divide by ad – bc.

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Finding A^{-1} with Gauss-Jordan

Three steps

1. Write the augmented matrix [A I].
2. Row-reduce until the left side is I.
3. The right side is now A^{-1}.

Quick checks

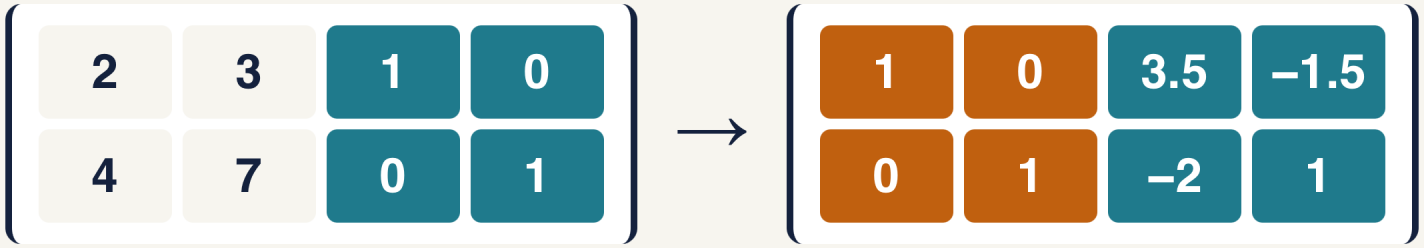
A zero row or fewer than n pivots means singular (no inverse).

Diagonally dominant matrices are invertible: each diagonal entry beats the rest of its row.

Triangular matrices: invertible if no diagonal entry is 0.

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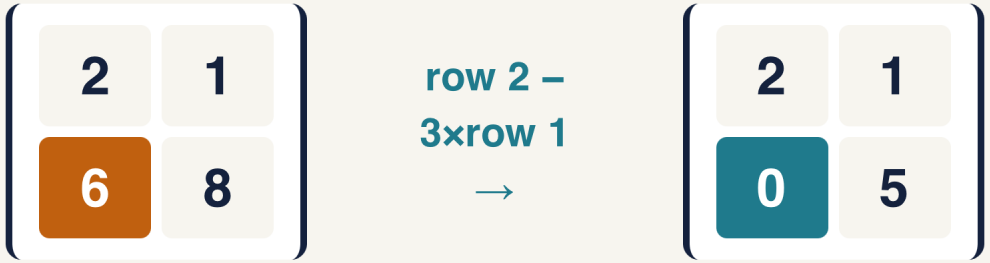
Visual example: finding A^{-1}



Start with [A I]. Row steps: row 2 – 2·row 1, then row 1 – 3·row 2, then divide row 1 by 2.

The left side becomes I (orange). The right side becomes A^{-1} (teal).

2.6 Elimination step by step



Goal: turn the orange 6 into 0. Subtract 3 times row 1 from row 2.

The result U is upper triangular (zeros below the diagonal).

The 3 is the multiplier. We keep it, because L is built from these multipliers.

A = LU: elimination in reverse

$$\begin{bmatrix} 2 & 1 \\ 6 & 8 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} \times \begin{bmatrix} 2 & 1 \\ 0 & 5 \end{bmatrix}$$

Subtract 3 × row 1 from row 2 to reach U. The multiplier 3 (orange) goes straight into L.

L is lower triangular with 1s on its diagonal. U is upper triangular with the pivots 2 and 5.

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2.6 Why triangular matrices are easy



L (teal) has zeros above the diagonal, so each equation only needs earlier answers: solve top to bottom.

U (orange) has zeros below the diagonal: solve bottom to top (back substitution).

Splitting A into L and U turns one hard problem into two easy staircases.

Solving Ax = b with L and U

Ax = b → Lc = b (forward) → Ux = c (backward)

Forward: Lc = b

L = [1 0; 4 1], b = (5, 21)

$c_1 = 5$

$4 \cdot 5 + c_2 = 21$, so $c_2 = 1$

Backward: Ux = c

U = [1 2; 0 1], c = (5, 1)

$x_2 = 1$

$x_1 + 2 \cdot 1 = 5$, so $x_1 = 3$

Factor once (about $n^3/3$ steps). Each new b only costs n^2 steps.

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Good to know about LU

A = LDU

Divide each row of U by its pivot.

The pivots go into a diagonal matrix D.

Zero pivot

Without row exchanges, A = LU needs nonzero pivots.

Otherwise we use a permutation matrix P.

Speed

Band matrices cost about $n \cdot w^2$ instead of $n^3/3$.

Most real matrices are sparse, so LU is fast.

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Key takeaways

- AB uses dot products; sizes $(m \times n)(n \times p)$ give $m \times p$.
- Multiplication is associative but not commutative.
- Blocks multiply like numbers.
- A^{-1} exists exactly when A has n pivots; $(AB)^{-1} = B^{-1}A^{-1}$.
- Gauss-Jordan turns $[A\ I]$ into $[I\ A^{-1}]$.
- $A = LU$: L stores the multipliers; solve $Lc = b$, then $Ux = c$.

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Practice exercises

1. Multiply [1 2; 3 4] by [0 1; 1 0].
2. Find the inverse of [1 2; 3 7] with the 2×2 formula.
3. Factor A = [1 1; 3 5] into L times U.
4. With that L and U, solve Ax = (2, 8): first Lc = b, then Ux = c.
5. If A and B are invertible, what is $(AB)^{-1}$?

Exercise answers

1. [2 1; 4 3]. The columns of A are swapped.
2. ad – bc = 7 – 6 = 1, so the inverse is [7 –2; –3 1].
3. L = [1 0; 3 1] and U = [1 1; 0 2].
4. c = (2, 2), then x = (1, 1).
5. $B^{-1}A^{-1}$: the order reverses.

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